

Project Proposal

Autonomous Multi-Agent Legal Contract Review System

Prepared by Muhammad Abuzar Ejaz

1. Executive Summary

The Multi-Agent Legal Contract Review System is an automated software tool designed to read, understand, and analyze complex legal documents. Instead of lawyers spending hours reading through pages of dense text, users simply upload a PDF of a commercial contract. The system uses a team of four specialized AI agents working together to extract the text, identify 41 specific types of legal clauses, and flag potential risks. It outputs a clean, structured JSON report that highlights problematic areas, provides a risk score from 1 to 10, and offers actionable negotiation tips.

Core Technical Workflow Loop (The “Wow” Factor)

The standout feature of this platform is the collaborative, multi-agent AI team. The workflow moves logically from a Parser Agent that cleans up raw text, straight to a Clause Classifier and Risk Analyzer, and finally to a Report Generator. This ensures that every risk score is backed by a specific, identified clause. It takes the messy, time-consuming process of manual legal review and turns it into a structured, automated pipeline that tells the user exactly what needs to be negotiated.

2. Problem Statement & Operational Context

Reviewing legal contracts is famously slow, expensive, and prone to human error. Finding a risky clause in a 50-page document is like finding a needle in a haystack. Lawyers, law firms, and startup founders currently spend countless hours manually checking for missing NDAs, unfair liability terms, or tricky payment rules. This project solves that problem by doing the heavy lifting instantly. It gives users immediate peace of mind by clearly highlighting what they are signing and what they need to push back on before closing a deal.

3. Why a Multi-Agent Pipeline Instead of a Single Chat Query

A natural question is why this needs a dedicated pipeline at all when a contract could simply be pasted into a chat-based assistant. For a single, one-off contract, a chat session can get close to the same result with no engineering effort. The justification for a dedicated system shows up once reliability, scale, and integration matter.

- **Structured, guaranteed output:** Claude in a chat gives back a well-written paragraph, but the format isn't guaranteed — ask twice and the shape of the answer can change. This pipeline forces every contract through the same JSON schema, so the output can be plugged into a dashboard, a database, or a law firm's case management software. A chat reply cannot be integrated anywhere.

- **Designed to run unattended, at scale:** A firm reviewing hundreds of contracts does not want to copy-paste each one into a chat window individually. This system is an automated backend — upload a document, receive structured JSON back. That is a product, not a conversation.
- **Specialization beats general-purpose for a fixed taxonomy:** The system always needs to recognize the same 41 clause types, labeled consistently. A general-purpose model asked to “find the risky parts” will phrase things differently each run. Fine-tuning on the CUAD dataset with QLoRA locks in consistent categorization, which matters when comparing risk scores across many contracts.
- **Traceability:** A risk score on its own is not useful to a lawyer. The pipeline requires every flagged risk to point to the exact page and paragraph behind it — an explicit extraction step that a single chat answer does not naturally guarantee.
- **Cost and control at scale:** Running thousands of contracts through a chat interface is not practical — there is no batch mode there. Running the pipeline on a self-hosted or API-based model (such as Groq) keeps latency and cost per document under control, which matters once this moves from a personal favor to a real product.

In short, a general-purpose assistant can perform each individual step well, but a chat session does not provide the reliability, structure, or scale needed for repeated, large-volume legal review. This system is best understood as that same underlying intelligence, engineered into a repeatable backend instead of a one-off conversation.

4. End-to-End Pipeline Architecture

The system is powered by four distinct AI agents that handle the document step-by-step:

- **Agent 1 (The Parser):** Pulls the raw text out of the uploaded PDF and cleans up any messy formatting so the other agents can read it clearly.
- **Agent 2 (The Classifier):** Scans the clean text to find and categorize 41 different types of important legal clauses (such as termination rules, payment details, and confidentiality agreements).
- **Agent 3 (The Risk Analyzer):** Evaluates the identified clauses to determine whether they are dangerous or unfair, assigning a risk score from 1 to 10 for the overall contract.
- **Agent 4 (The Report Generator):** Takes all of this information and creates a final, easy-to-read structured JSON report — including the parties involved, exactly where to find each clause in the document, and practical suggestions for negotiating the risky parts.

System Architecture Diagram

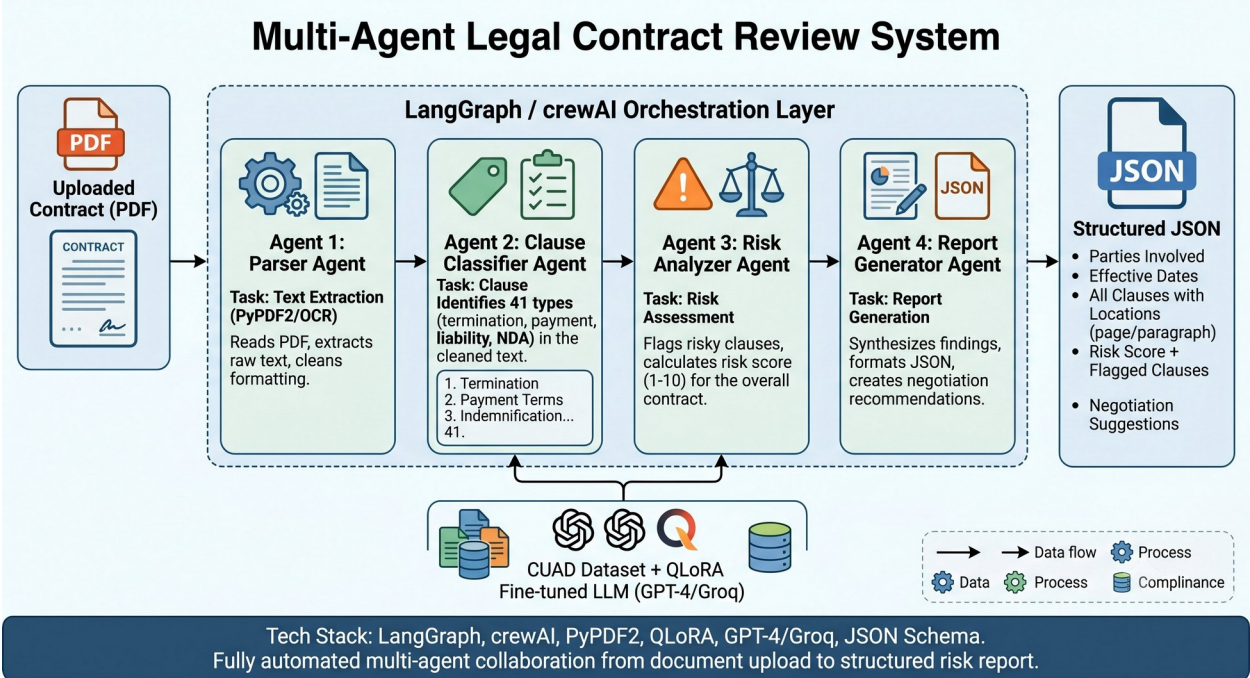


Figure 1: Multi-Agent Legal Contract Review System — end-to-end pipeline from PDF upload through the LangGraph/crewAI orchestration layer to the structured JSON output.

5. Specialized Integration Architecture

System Layer	Integrated Services	Functional Objective
Document Processing	PyPDF2, OCR Tools	Extracts and cleans raw text from uploaded contract PDFs.
Model & Fine-Tuning Layer	GPT-4 / Groq, LoRA / QLoRA	Handles deep understanding of legal language, customized for the legal domain.
Agent Orchestration	LangGraph / crewAI	Manages step-by-step collaboration between the four AI agents.
Data & Output Layer	CUAD Dataset, JSON Schema	Guides the system's knowledge base and formats the final negotiation report.

6. Target Audience & Technical Implementation

This platform is built for law firms, startup founders, and enterprise legal teams who need to review commercial contracts quickly and safely. To bring this to life, I will build the core

architecture using Python and LangGraph for the multi-agent teamwork. Rather than relying entirely on generic models, the system leverages targeted QLoRA fine-tuning techniques applied to the CUAD dataset, ensuring the underlying Transformer architectures are highly specialized for legal clause extraction without excessive computational overhead.

7. Planned Enhancements (Future Work)

The following enhancements are not part of the initial build but are identified as natural next steps once the core four-agent pipeline is stable:

- **Confidence Scoring:** Each flagged clause will carry a confidence value alongside its risk score, so low-confidence flags can be routed for human review instead of being presented with false certainty — important given the cost of a wrong call in a legal context.
- **Human-in-the-Loop Feedback:** A simple interface for a lawyer to mark a flagged clause as a false positive, with that feedback logged and periodically used to refine the fine-tuned model. This allows the system's accuracy to improve with real usage rather than staying static.
- **Benchmark Evaluation:** Formal evaluation of the Classifier and Risk Analyzer agents against CUAD's labeled test set (precision/recall on clause detection, agreement with expert risk ratings) to quantify accuracy rather than relying on demonstration alone.